



ABSTRACT

This project presents an AI-Powered Solar Panel Cleaning Scheduler that autonomously detects dust accumulation and efficiency drops, and dynamically schedules cleaning operations. The system integrates a NodeMCU (ESP8266) microcontroller, RTC DS3231, current sensor, relay module, and a manual push button — all powered by a 10V solar battery. An AI-based decision module evaluates three trigger conditions: dust detection (current < 0.28 A), daily scheduled cleaning at 10:00 AM via RTC, and manual button override. The system uploads real-time current values and cleaning status to ThingSpeak IoT cloud every 20 seconds via Wi-Fi, enabling remote performance monitoring. Experimental results show panel efficiency restored from 61.3% to 95.8% after cleaning, with cleaning frequency reduced by 50% compared to fixed-interval methods.

TEAM MEMBER DETAILS

1. B. Raghavendra Reddy VTU21873
Ph no: 6301493043
Mail: vtu21873@veltech.edu.in
2. Y. Gavaskar Babu VTU22974
Ph no: 8500944373
Mail: vtu22974@veltech.edu.in
3. I. Atchi Reddy VTU21872
Ph no: 9515809410
Mail: vtu21872@veltech.edu.in

INTRODUCTION

The global demand for sustainable clean energy has accelerated deployment of solar photovoltaic (PV) systems. However, their long-term performance is severely affected by soiling — dust deposition on panel surfaces. Studies show soiling causes power losses of 5–30% depending on local climatic conditions and maintenance frequency.

Traditional periodic or manual cleaning methods are inefficient and waste resources. AI and IoT technologies enable predictive and adaptive cleaning schedules. Sensors continuously collect real-time data enabling automated decision-making without human intervention.

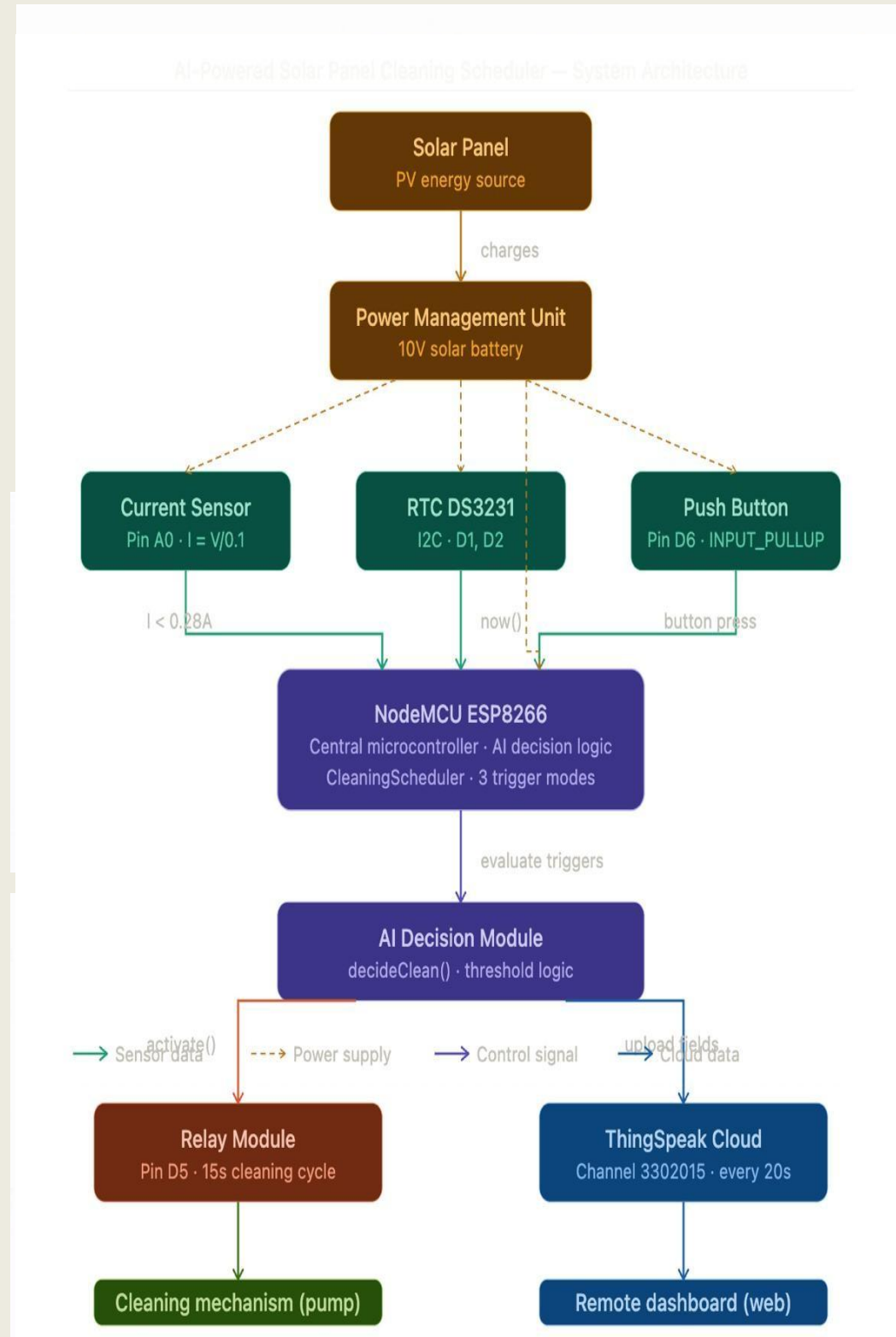
This project combines a current-sensor-based dust detection mechanism, RTC-based scheduling, relaycontrolled cleaning, and ThingSpeak IoT monitoring into a fully self-sustaining, intelligent solar maintenance system.

METHODOLOGIES

The system follows a structured approach for detecting dust and automating cleaning:

1. Sensing & Data Acquisition: Current sensor on pin A0 reads analog voltage every 2 seconds. Converted via $I = V/0.1$. Dust flagged when $I < 0.28$ A.
2. Real-Time Scheduling (RTC DS3231): Provides timestamps via I²C (D1, D2). Triggers mandatory daily cleaning at 10:00 AM. Resets cleaned Today flag at midnight.
3. AI-Based Decision Logic: Cleaning Scheduler evaluates 3 triggers: dust detection, scheduled time, manual button press. Prevents duplicate cycles via status flags.
4. Relay-Controlled Cleaning: On any valid trigger, relay (D5) activates for exactly 15 seconds then deactivates. Logs completion and resets cleaning flag.
5. Manual Override: Physical push button on D6 (INPUT_PULLUP) with 500ms debounce allows instant user-initiated cleaning independent of sensors.
6. IoT Cloud Monitoring: NodeMCU uploads current value (Field 1) and cleaning status (Field 2) to ThingSpeak channel 3302015 every 20 seconds via Wi-Fi.

ARCHITECTURE DIAGRAM



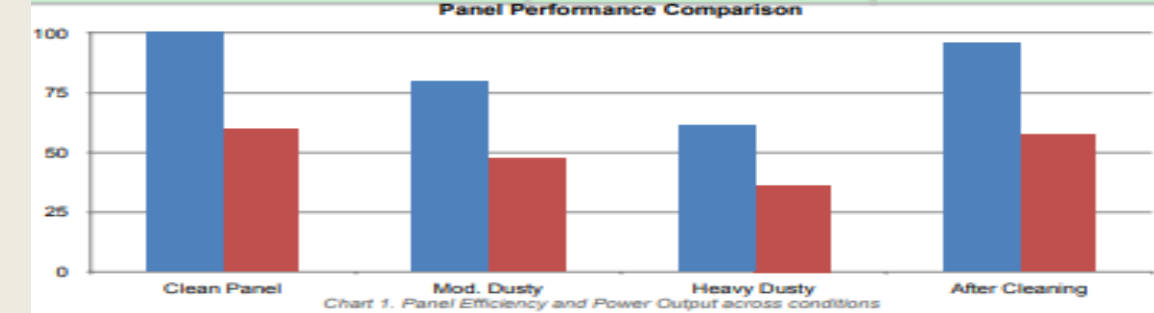
RESULTS

Table 1. Solar Panel Performance Before and After Cleaning

Condition	Voltage (V)	Current (A)	Power (W)	Efficiency (%)
Clean Panel	18.6	3.2	59.52	100%
Moderately Dusty	16.9	2.8	47.32	79.5%
Heavily Dusty	15.2	2.4	36.48	61.3%
After Cleaning	18.4	3.1	57.04	95.8%

Table 2. Cleaning Frequency Analysis Over 30 Days

Method	No. of Cleanings	Saving
Fixed Interval Method	12 cleanings	—
Proposed AI System	6 cleanings	50% ↓



CONCLUSIONS

- The proposed AI-powered solar panel cleaning scheduler successfully demonstrates automated, intelligent, and selfsustaining PV maintenance
- Panel efficiency restored from 61.3% to 95.8% after automated cleaning
 - Cleaning frequency reduced by 50% vs. fixed-interval methods
 - System operates fully off-grid on solar battery power
 - ThingSpeak enables real-time remote IoT monitoring

PUBLICATION

- Submitted-Published

SUPERVISOR

1. Name: Dr. Peer Mohamad Asst.prof,S.G
2. Ph no: 8189871231
3. Mail: peermohamad@veltech.edu.in